

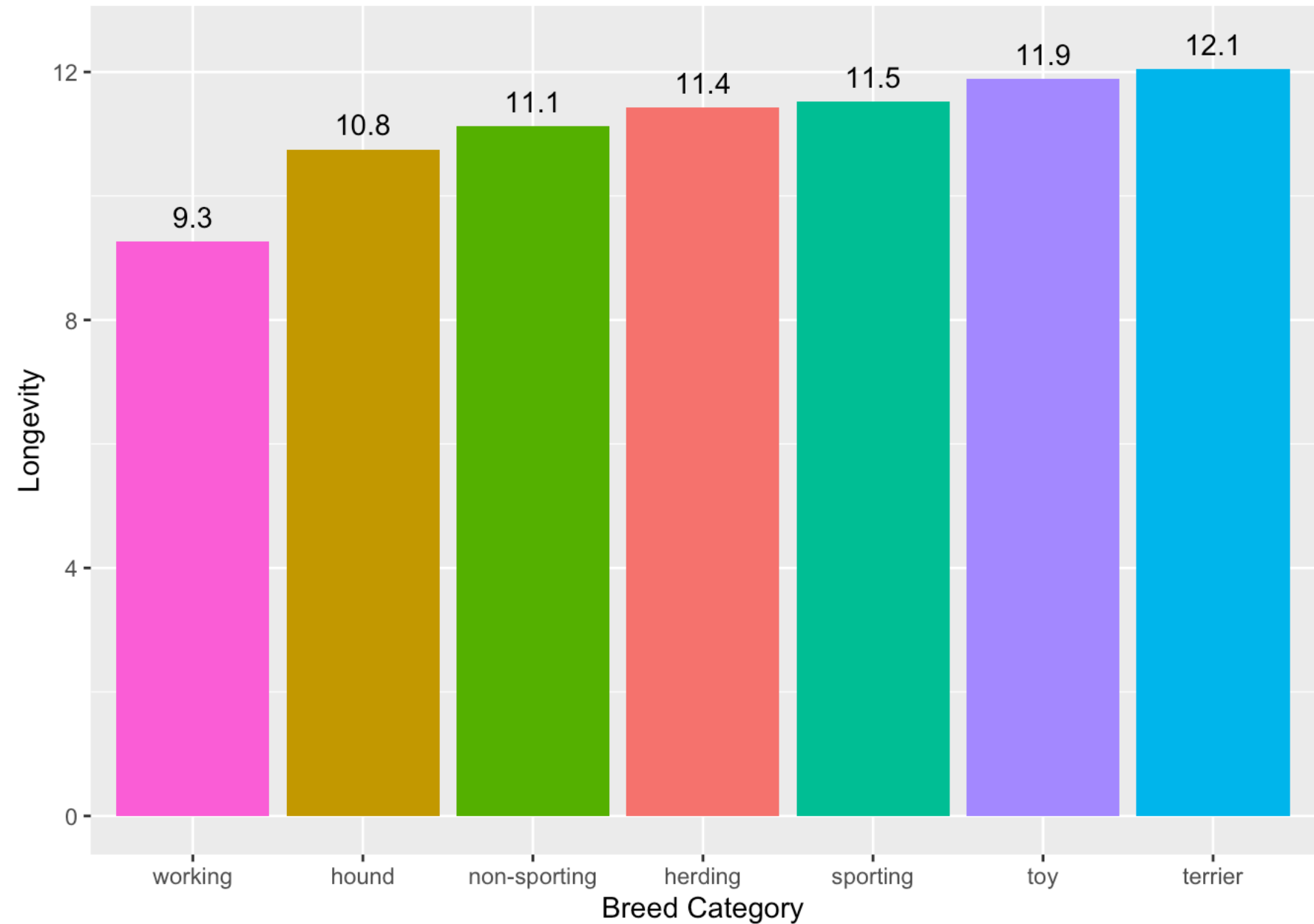
# Breed Longevity

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This Project studies the average longevity of seven different breed categories. Breeds are sorted according to the American Kennel Club (AKC), and the raw data is also from the AKC. The breeds are sorted into herding breeds, terrier breeds, sporting breeds, toy breeds, working breeds, non-sporting breeds, and hound breeds. Individual breeds are sorted according to their breed type in the table below:

| Dog Breeds by Type    |                             |                            |                               |                      |                 |                     |
|-----------------------|-----------------------------|----------------------------|-------------------------------|----------------------|-----------------|---------------------|
| herding               | terrier                     | sporting                   | toy                           | working              | non-sporting    | hound               |
| Border Collie         | Border Terrier              | Brittany                   | Papillon                      | Siberian Husky       | Lhasa Apso      | Dachshund           |
| Australian Cattle Dog | Cairn Terrier               | Welsh Springer Spaniel     | Affenpinscher                 | Samoyed              | Poodle          | Whippet             |
| Shetland Sheepdog     | Miniature Schnauzer         | English Cocker Spaniel     | Chihuahua                     | Doberman Pinscher    | Bichon Frise    | Pharaoh Hound       |
| Australian Shepherd   | Australian Terrier          | Cocker Spaniel             | Pomeranian                    | Giant Schnauzer      | Tibetan Spaniel | Beagle              |
| Briard                | West Highland White Terrier | English Springer Spaniel   | Maltese                       | Rottweiler           | Tibetan Terrier | Basset Hound        |
| Belgian Tervuren      | Bedlington Terrier          | Pointer                    | Shih Tzu                      | Newfoundland         | Boston Terrier  | Basenji             |
| Bouvier des Flandres  | Norfolk Terrier             | German Shorthaired Pointer | Yorkshire Terrier             | Akita                | Dalmatian       | Saluki              |
| Pembroke Welsh Corgi  | Staffordshire Bull Terrier  | Labrador Retriever         | Brussels Griffon              | Bernese Mountain Dog | French Bulldog  | Greyhound           |
| German Shepherd       | Dandie Dinmont Terrier      | Irish Setter               | Pug                           | Boxer                | Chow Chow       | Afghan Hound        |
| Old English Sheepdog  | Scottish Terrier            | Golden Retriever           | English Toy Spaniel           | Alaskan Malamute     | Bulldog         | Rhodesian Ridgeback |
|                       | Kerry Blue Terrier          | Chesapeake Bay Retriever   | Cavalier King Charles Spaniel | Bullmastiff          |                 | Borzoi              |
|                       | Bull Terrier                | Gordon Setter              | Italian Greyhound             | Mastiff              |                 | Irish Wolfhound     |
|                       |                             | English Setter             | Pekingese                     | Great Dane           |                 | Bloodhound          |
|                       |                             | Flat-Coated Retriever      |                               | Saint Bernard        |                 |                     |
|                       |                             | Clumber Spaniel            |                               |                      |                 |                     |

The raw data is from the AKC's Canine Inherited Disorders Database, while the data set I used to create these graphics is from David McCandless at [informationisbeautiful.net](#). For this project, I averaged the average lifespans among breeds in each AKC category, with the result being the average lifespan of each AKC category. I then present this data visually in a bar graph. This data is a useful tool for someone looking to get a dog, but worried about potential health issues. This would be an especially useful research starting point, as potential owners can further examine why certain types have shorter lifespans, and see a further breakdown by breed. It is also a useful for tool for breeders and veterinary scientists looking to identify which breed groups live the longest, allowing them to determine if there are possible genetic factors that contribute to longevity across breeds within their types.



As one can see, the terrier group has the longest lifespan. From the data set, one can see that this lifespan is averaged across breeds at 12.1 years, with the minimum at 9.4, belonging to the Kerry blue terrier, and the maximum at 13.5, belonging to the Bedlington terrier. The lowest longevity belongs to the working breeds. This lifespan is averaged across breeds at 9.3 years, with the minimum at 6.5 belonging to the mastiff, and the maximum at 10.7, belonging to the Alaskan malamute. The working breeds having the lowest average lifespans aligns with common wisdom of dog breed longevity, as this group contains some of the largest dog breeds, such as great danes, saint bernards, and great pyrenees. Likewise, it is expected that the toy and terrier breeds have the longest lifespans, as these contain smaller breeds, such as the chihuahua (toy), and the Scottish terrier (terrier). Perhaps terriers have longer lifespans because they are bred with working characteristics in mind, while toy breeds are often bred with aesthetic characteristics in mind. However, the difference between the two averages is small, only the matter of two decimal points. Therefore, the difference could simply be a result of random error, and it is worth seeing if another data set ranks toys and terriers similarly.

If I were a veterinary scientist, I would study what specifically gives the terrier breeds have the greatest longevity prospects. Is there something in their genetics that makes them less predisposed to certain illnesses, or is it a matter of environment? Perhaps these breeds are more likely to go to wealthy families, where there is abundant veterinary care available. It would also be useful to study why exactly working breeds have such low longevity prospects. Again, it could be genetic, such as being predisposed to certain illnesses, or it could be environmental, such as being more likely to be abandoned.

While this data is a useful starting point, prospective dog owners and veterinary scientists should examine further if and how these differences are significant. How similar are breeds within each category, really? Do these breeds have similar health predispositions? Asking these questions will help clarify what it is that causes certain breeds to have lower life expedienicies.